NW

NEXWAVE TELCO

Customer Churn Prediction

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Presentation Outline

01 Business Problem Understanding

Context, problem statement, stakeholders, goal, analytical approach, and metric evaluation.

02 Data Understanding & EDA

Dataset overview, attribute information, target distribution, feature insights, and correlations.

03 Data Cleaning & Preparation

Duplicate removal, feature engineering, encoding, train-test split, and SMOTE resampling.

04 Modeling & Evaluation

Baseline model comparison, SMOTE + hyperparameter tuning, threshold optimization, feature importance.

05 Business Impact

Quantified cost savings and ROI from deploying the churn prediction model.

06 Conclusion & Recommendation

Key findings, strategic business actions, and technical next steps.

01

Business Problem Understanding

Context | Problem Statement | Goals | Stakeholders | Analytical Approach | Metric Evaluation

Business Context

NW

NEXWAVE TELCO

Telecommunications Service Provider

- Provides internet, WiFi, and telephone services
- Offers individual and bundled service packages
- Serves thousands of customers across the country
- Revenue dependent on monthly subscription fees

Acquiring a new customer costs 5-25x more than retaining an existing one

Problem Statement

NexWave Telco cannot identify which customers are likely to churn before they actually leave, making proactive retention efforts impossible.

Goals

Build an ML model to predict customer churn so the retention team can intervene proactively, prioritize high-risk customers, and reduce revenue loss.

Stakeholders

NexWave Telco Senior Management and Customer Retention Team.

Analytical Approach: Binary Classification using Machine Learning

Metric Evaluation

Confusion Matrix

	Predicted: No Churn	Predicted: Churn
Actual: No Churn	**True Negative (TN)** Correctly predicted Not Churn	**False Positive (FP)** Wrongly predicted Churn
Actual: Churn	**False Negative (FN)** Missed churners — HIGH COST	**True Positive (TP)** Correctly predicted Churn

FP **False Positive (FP)**

Predicting churn when the customer stays. Cost: unnecessary retention spend (vouchers, discounts).

FN **False Negative (FN)**

Missing a churning customer. Cost: permanent revenue loss — no chance to retain the customer.

F2 **Primary Metric: F2-Score**

Weights Recall 2x more than Precision — penalizes FN heavily. Supporting: Precision, Recall, F1-Score, ROC-AUC.

02

Data Understanding & Exploratory Analysis

Dataset Overview | Target Distribution | Feature Insights | Correlations

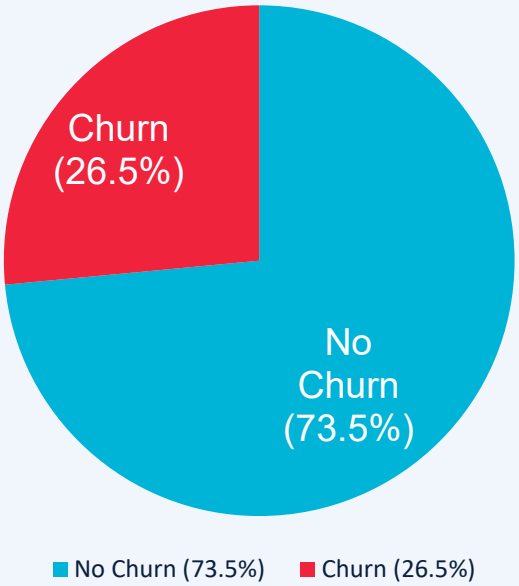
Data Understanding

4,930	11	0	26.5%
Initial Records	Features	Missing Values	Churn Rate

Feature	Type	Description	Range/Values
Dependents	Categorical	Customer has dependents	Yes / No
tenure	Numerical	Subscription length in months	0 – 72 months
OnlineSecurity	Categorical	Online Security service status	Yes / No / No internet
OnlineBackup	Categorical	Online Backup service status	Yes / No / No internet
InternetService	Categorical	Type of internet service	DSL / Fiber optic / No
DeviceProtection	Categorical	Device Protection status	Yes / No / No internet
TechSupport	Categorical	Tech Support service status	Yes / No / No internet
Contract	Categorical	Contract type (ordinal)	Month-to-Month / 1yr / 2yr
PaperlessBilling	Categorical	Uses paperless billing	Yes / No
MonthlyCharges	Numerical	Monthly bill amount (USD)	$18.80 – $118.65
Churn	**Target**	**Customer churned**	**Yes / No**

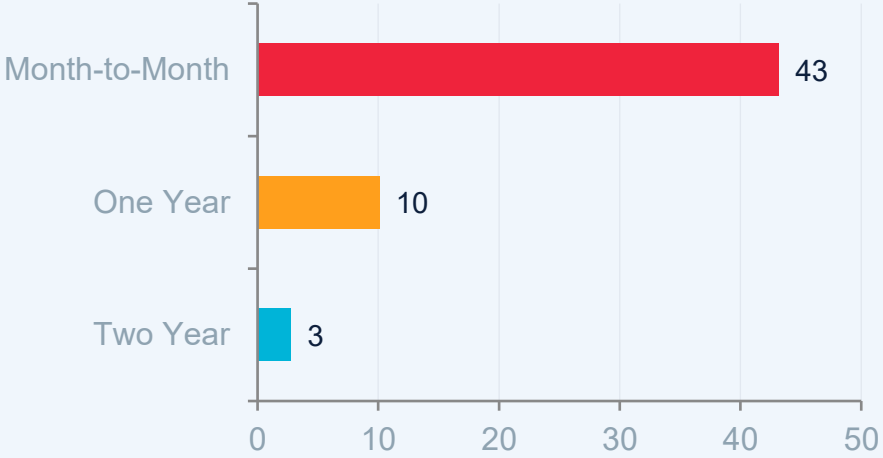
EDA: Class Imbalance & Key Churn Drivers

Target Distribution



Class Imbalance Ratio: 2.77:1 — SMOTE required

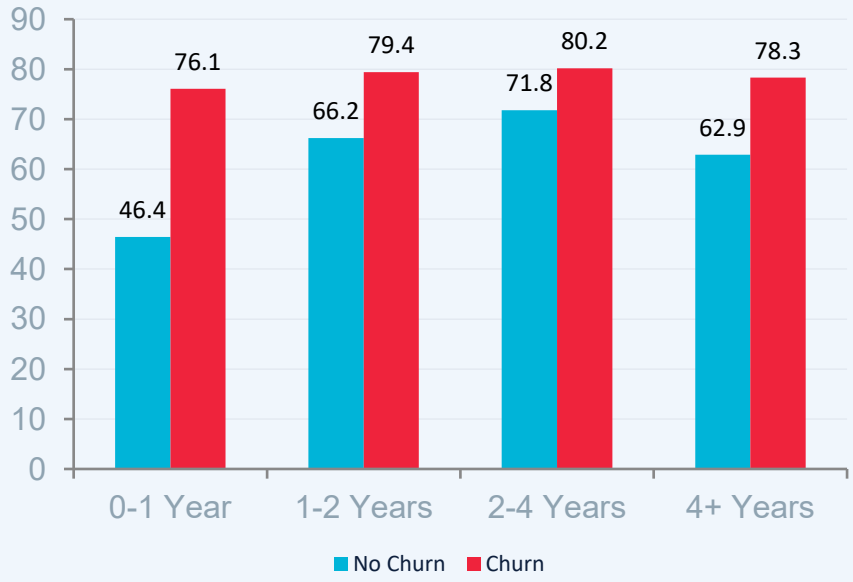
Churn Rate (%) by Contract Type



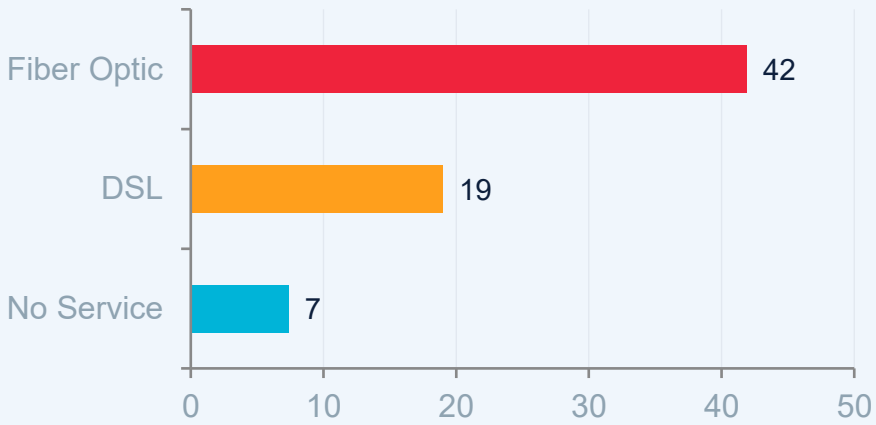
Key Insight: 74.6% of churned customers are on Month-to-Month contracts. Longer contracts are strongly correlated with lower churn — driving contract upgrades is critical.

EDA: Tenure & Monthly Charges vs Churn

Avg. Monthly Charges by Tenure Group



Churn Rate by Internet Service



Key Insight: Churned customers consistently pay higher monthly charges across all tenure groups. New subscribers (0-1 year) who churn pay ~64% more than those who stay.

03

Data Cleaning & Preparation

Data Cleaning | Feature Engineering | Encoding | Splitting | SMOTE Resampling

Data Cleaning & Feature Engineering

Data Cleaning Pipeline

Raw Data
4,930 rows
11 columns

Remove Duplicates
−77 rows
(1.56%)

Check Missing
0 nulls
No imputation

Anomaly Check
8 tenure=0
Kept (valid)

Clean Data
4,853 rows
11 columns

Feature Engineering — 4 New Features Created

tenure_group
Ordinal

Segments customers into loyalty tiers: 0-1yr, 1-2yr, 2-4yr, 4+yr — captures non-linear churn behavior by lifecycle stage.

has_internet
Binary Flag

Flags whether the customer has any internet service — consolidates DSL/Fiber into a single binary predictor.

num_addon_services
Numerical

Counts the number of additional services subscribed (0-4). More services = higher engagement = lower churn risk.

charge_category
Ordinal

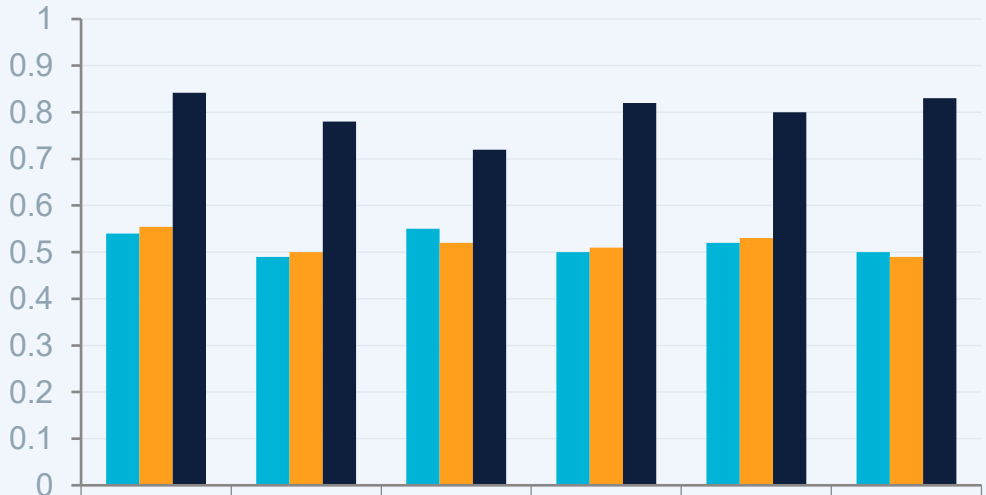
Buckets MonthlyCharges into Low/Medium/High/Very High — simplifies pricing tiers for the model.

04

Modeling & Evaluation

Baseline Models | SMOTE | Hyperparameter Tuning | Threshold Optimization | Feature Importance

Baseline Model Comparison (6 Models, 5-Fold CV)



	Logistic Regression	K-Nearest Neighbors	Decision Tree	Random Forest	AdaBoost	XGBoost
Recall	0.54	0.49	0.55	0.5	0.52	0.5
F2-Score	0.5542	0.5	0.52	0.51	0.53	0.49
ROC-AUC	0.842	0.78	0.72	0.82	0.8	0.83

Recall F2-Score ROC-AUC

Best Model Selected

Logistic Regression

F2-Score: **0.5542**
Recall: **0.5388**
ROC-AUC: **0.8420**

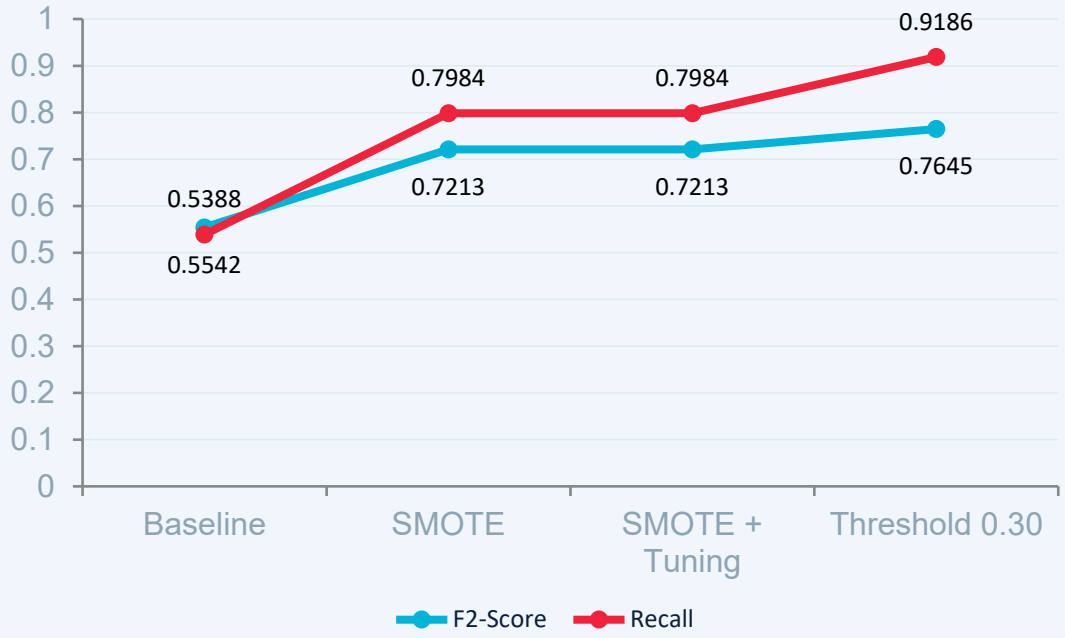
Why Logistic Regression?

- Highest F2-Score in baseline
- Highest ROC-AUC (0.842)
- Strong interpretability via coefficients
- Fast training + scalable

Improvement Strategy

1. SMOTE — address class imbalance
2. GridSearchCV — hyperparameter tuning
3. Threshold Optimization — max F2

Model Improvement: SMOTE + Tuning + Threshold



Final Model Metrics (Threshold = 0.30)

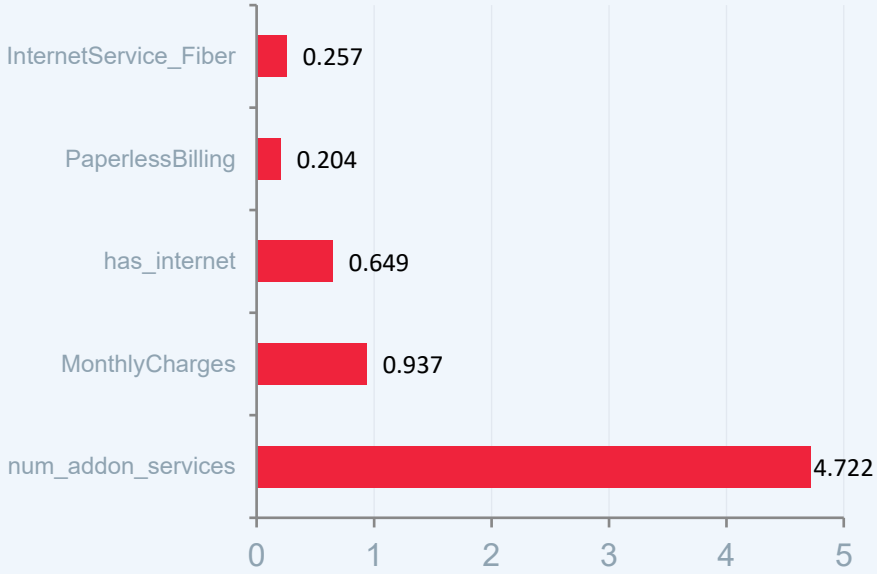
ROC-AUC	
F2-Score	**0.7645**
Recall	**0.9186**
Precision	**0.4575**
F1-Score	**0.6108**
Accuracy	**0.6890**

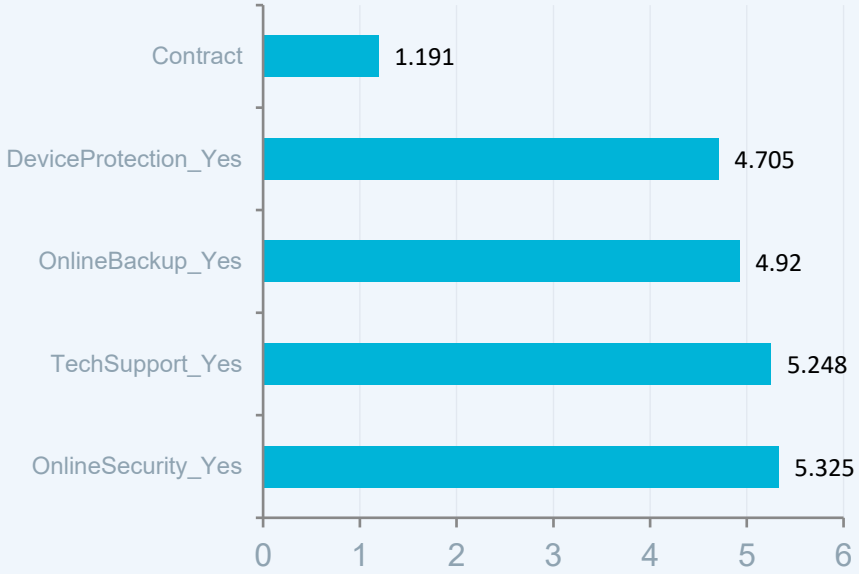
Tuned Params: C = 20, penalty = L2 (GridSearchCV, 5-Fold CV)
Threshold optimized to maximize F2-Score

SMOTE Impact:

Before SMOTE: {No Churn: 2,852 | Churn: 1,030}
After SMOTE: {No Churn: 2,852 | Churn: 2,852} → perfectly balanced training data

Feature Importance – Logistic Regression Coefficients





Churn Risk Increasing Factors (positive coefficients)

Churn Risk Decreasing Factors (negative coefficients)

Model Insight: OnlineSecurity and TechSupport are the strongest protective factors — each subscription reduces churn log-odds by ~5.3 and ~5.2 respectively. Conversely, having no add-on services (num_addon_services) strongly drives churn risk. Contract type also plays a major role — longer contracts significantly reduce churn probability.

05

Business Impact

Cost Savings Analysis | Without Model vs With Model | ROI Estimation

Business Impact: Cost-Benefit Analysis

Cost Assumptions (per customer):

False Positive (FP — unnecessary retention offer): $10.60 | False Negative (FN — missed churner, customer lost): $53.00 (avg. customer acquisition cost)

WITHOUT ML MODEL

	Pred: No Churn	Pred: Churn
Act: No Churn	TN = 0	FP = 713
Act: Churn	FN = 0	TP = 258

Scenario: No model — treat ALL customers as potential churners

Total Estimated Cost:

713 × $10.60 = $7,557.80

WITH ML MODEL

	Pred: No Churn	Pred: Churn
Act: No Churn	TN = 435	FP = 278
Act: Churn	FN = 21	TP = 237

Model correctly identifies 91.9% of churners for timely intervention

Total Estimated Cost:

(278×$10.60)+(21×$53)=$4,060.80

06

Conclusion & Recommendation

Summary of Findings | Strategic Business Recommendations | Technical Next Steps

Conclusion

01 Dataset Overview

4,853 clean customer records with 11 features. No missing values; 77 duplicates removed. Class imbalance of 2.77:1 (No Churn:Churn) was addressed using SMOTE.

02 Key Churn Drivers

Month-to-Month contracts (~43% churn rate), short tenure (0-12 months), Fiber Optic service (~42% churn), high monthly charges, and absence of add-on services (OnlineSecurity, TechSupport) are the primary churn predictors.

03 Model Performance

Final model: Logistic Regression + SMOTE + Hyperparameter Tuning (C=20, L2) + Threshold 0.30. Achieves F2-Score of 0.7645 and Recall of 91.9% — capturing 9 in 10 churning customers.

04 Business Value

Deploying this model reduces estimated retention cost from $7,557 to $4,061 per test cycle — a 46% cost reduction. The model enables proactive, targeted interventions instead of blanket campaigns.

Business Recommendations

HIGH

Proactive Retention for Month-to-Month Customers

Monthly contract customers

Use the ML model to identify high-risk monthly subscribers and trigger personalized retention offers (loyalty discounts, free service upgrades) before churn occurs.

HIGH

Contract Upgrade Incentive Program

Customers with tenure < 12 months

Offer discounted annual or two-year contracts in the first 3-6 months. New customers who lock into longer contracts have dramatically lower churn rates (~3% for 2-year contracts).

MEDIUM

Bundle Additional Services (Security + TechSupport)

Customers without add-on subscriptions

Promote bundled packages that include OnlineSecurity and TechSupport. These are the two strongest retention factors per the model — each reduces churn risk significantly.

MEDIUM

Loyalty & Reward Program

Long-term customers (tenure > 24 months)

Reward loyal customers with exclusive benefits to reinforce retention behavior and deter competitive switching — especially for Fiber Optic users where churn risk from pricing is high.

LONG TERM

Fiber Optic Pricing Review vs. Competitors

All Fiber Optic subscribers

Fiber optic users exhibit a 41.9% churn rate, likely driven by high prices relative to market competition. A competitive pricing audit and tiered pricing strategy are recommended.

LONG TERM

Deploy ML Early Warning System

Entire customer base

Integrate the churn model into the CRM system to generate monthly churn risk scores for all customers. Enable the retention team to work from a prioritized intervention list.

Technical Recommendations & Model Limitations

01 Enrich Feature Set

Add customer satisfaction scores, complaint frequency, payment history, and NPS scores. These behavioral signals are strong churn predictors not yet captured in the current dataset.

02 Quarterly Model Re-evaluation

Customer behavior evolves over time. The model should be retrained quarterly using fresh data to prevent concept drift and maintain prediction accuracy.

03 A/B Testing for Retention Campaigns

Test the effectiveness of retention interventions on ML-predicted high-risk customers vs. a control group to quantify actual retention lift and ROI.

04 Dynamic Threshold Adjustment

Adjust the decision threshold based on the ratio of retention cost to customer lifetime value. As cost assumptions change, the optimal threshold should be recalibrated accordingly.

Model Limitations

- Lower precision (45.8%) — some non-churners receive unnecessary retention offers.
- Only 11 original features — limited view of full customer behavior.
- No temporal data — cannot capture customer behavior trends over time.
- Synthetic minority data (SMOTE) may not perfectly represent real churn patterns.
- Model trained on one snapshot — performance may degrade over time without retraining.

Despite limitations, the model delivers immediate business value and provides a scalable foundation for a production-grade churn prediction system.

Thank You

91.9%	0.76	46%	0.84
Model Recall	F2-Score	Cost Reduction	ROC-AUC

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